MIL-L-44089A
27 March 1985
SUPERSEDING
MIL-L-44089
20 September 1982

MILITARY SPECIFICATION

LINER, NIGHT CAMOUFLAGE PARKA: DESERT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This document covers a button-in insulating liner for the night camouflage desert parka The liner is made of nylon fabric quilted to a polyester batting material

1.2 Classification. The liner shall be of one type in the following sizes as specified (see 6.2).

Schedule of sizes

X-Small
Small
Medium
Large
X-Large

2 APPLICABLE DOCUMENTS

2.1 Government documents Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-280	-	Thread, Gimp, Cotton, Buttonhole
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-508	-	Cloth, Oxford, Nylon, 3 Ounce
MIL-B-41826	-	Batting, Synthetic Fibers· Polyester (Quilted and Unquilted)
MIL-T-43548	-	Thread, Polyester, Cotton-Covered and Rayon-Covered
MIL-T-43566	-	Tape, Textile, Cotton, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

2.2 Other publications Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

COLOR ASSOCIATION OF THE UNITED STATES INC.

Department of Defense Standard Shades for Sewing Threads

(Application for copies should be addressed to the Color Association of the United States Inc , 24 East 38th Street, New York, NY 10016)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3 1 Guide samples Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3 2 Materials (see 6.4).

3.2.1 Basic material. The basic material shall be a quilted polyester batting, conforming to type IV, class 1, style a of MIL-B-41826.

3 2 2 Facing and binding The material for facings, neck buttonhole reinforcement and binding shall be nylon oxford cloth, 3-ounce, Olive Green 106, conforming to type I, class 2 of MIL-C-508.

3.2.3 Tape. The tape for buttonhole tabs shall be type I, class 3, 1-inch width, dyed Olive Drab 7 conforming to MIL-T-43566.

3 2.4 Thread. The thread shall be cotton-covered, rayon-covered, polyester thread, dyed Olive Drab S-1, C A. 66022, ticket numbers 50, 2 or 3 ply and 70, 2 or 3 ply conforming to MIL-T-43548.

3 2.4.1 Colorfastness. The dyed thread shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering.

3.2 5 Gimp. The cotton gimp for reinforcing buttonholes shall be dyed Olive Drab S-1, C.A. 66022, size No. 8, conforming to type I or II of V-T-280.

3 2.5.1 Colorfastness. The dyed gimp shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed gimp shall show "good" fastness to laundering.

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3 2 6 Labels Each liner shall have an identification label, a size label and an instruction label conforming to DDD-L-20, except label color shall be Medium Green, Cable Nos. 70034, 70130, or 70131 (see 2 2) The fastness to laundering requirements shall apply

3 2 6 1 Identification label. The identification label shall conform to type VI, class 1

3 2 6 2 Size label The size label shall conform to type VI, class 2 and contain the following information for each respective size·

X-Small

Chest· Under 33 inches
Stock No.
NATO Size· 7080/7484

Small

Chest. From 33 to 37 inches
Stock No
NATO Size 7080/8494

Medium

Chest. From 37 to 41 inches
Stock No
NATO Size· 7080/9404

Large

Chest From 41 to 45 Inches
Stock No
NATO Size: 7080/0414

X-Large

Chest From 45 inches and upward
Stock No
NATO Size: 7080/1424

3 2.6 3 Instruction label The instruction label shall conform to type VI, class 3 and contain the following information:

LINER, NIGHT CAMOUFLAGE PARKA: DESERT

1. Button into parka for extra warmth
2 Dry liner and parka separately when damp or wet
3 Avoid contact with heated stoves, open fires, or other heat sources; fabric may fuse.
4 Wash by hand or machine in warm water with mild soap or detergent, rinse thoroughly and drip or machine dry. DO NOT DRY CLEAN.

DO NOT REMOVE LABEL

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3.2.6.4 Combination size and identification label. The identification label (class 1) and the size label (class 2) may be combined into one label conforming to type VI, classes 1 and 2 combined.

3.2.6 5 Combination size, identification and instruction label. The identification label (class 1), size label (class 2) and instruction label (class 3) may be combined into one label conforming to class 14.

3 3 Patterns. Standard patterns which provide 3/8 inch seam allowance for all seams except neck buttonhole reinforcement seams, will be furnished by the Government The standard patterns shall not be altered in any way and shall be used as a guide for cutting the contractor's working patterns only. The working patterns shall be identical to the standard patterns.

3 3.1 List of pattern parts. The components of the liner shall be cut from the materials specified below in accordance with the pattern parts indicted.

Material	Pattern nomenclature	Cut parts
Quilted polyester batting	Front 1/	2
	Back 1/	1
	Top sleeve	2
	Undersleeve	2
Cloth, nylon, oxford	Neck buttonhole reinforcement	1

1/ Fronts and back may be cut as one piece deducting seam allowances at side seams.

3.4 Construction. The construction shall conform in all respects to the requirements specified in table I and herein Figure 1 is furnished solely for guidance and information Should variation from this document appear in figure 1, this document shall govern

3 4.1 Stitches, seams, and stitching. All stitches, seams and stitching except bartacking shall conform to FED-STD-751 The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. Where two or more methods of seams or stitch types are given for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) may be on the outside or inside of the liner, except where seam type LSbm-3 is used, the looper shall be considered on the inside of the folded seam. The side of the liner toward the body, when worn, shall be considered the inside. The guides and knives on safety and overedging machines shall be set to trim only the ravelled ends of the fabric.

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3.4.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching Ends of continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn The lock shall be imbedded in the materials sewn.

3.4.1 1 1 Repairs of type 301 stitching.

3.4.1.1 1 1 Quilted batting stitching. Broken stitches of the quilting pattern may be repaired by stitching back of each end of break not less than 1 inch.

3.4.1.1.1 2 Liner fabrication stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.1 2 Type 401, 503, 504, 515, 516 and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching or excessively tight stitching resulting in puckering of the material sewn. All repairs shall be in accordance with 3 4.1.1.1.2.a and 3.4 1 1.1.2.b. Repairs to stitch type 401 may be done with stitch type 301.

3 4 2 Buttonhole The buttonholes shall be eyelet end, taper bar type, worked over gimp (see 3 2.5) with not less than four tacking stitches at bar end (not counting cross over stitch). The purling shall be on the facing side and reinforcement side. The finished cut lengths shall be 3/4 to 7/8 inches. The buttonholes shall be clean cut with the stitching securely caught in the fabric. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

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3.4.3 Bartacking. Unless otherwise specified, bartacking shall be as follows.

Length	Width	Stitches per bartack
1/2 inch ± 1/16 inch	1/8 inch ± 1/32 inch	28
3/4 inch ± 1/16 inch	1/8 inch ± 1/32 inch	42

Bartacking shall be free from thread breaks and loose stitching.

3 4 4 Automatic stitching. Automatic stitching may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.4.5 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3 4 6 Repairs. Repairs such as mends, darns, or patches are not permitted on the liner

3.5 Manufacturing operations requirements The liners shall be manufactured in accordance with all operation requirements specified in table I. The contractor is not required to follow the exact sequence of operations. Additional basting or holding stitching may be used providing the threads are removed or are not visible on the finished liner.

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NO	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
*	a. Cut the liner parts in strict accordance with patterns furnished, which show size, directional lines (warp and filling direction) and notches for proper assembly of parts. Fronts and back may be cut in warp or filling direction except that fronts and backs must be the same direction in each liner.						
	NOTE: Care shall be exercised when cutting the basic material and handling the cut parts in that the ends of the quilting stitching have a tendency to run back.						
	b. Cut the nylon oxford for the neck buttonhole reinforcement in accordance with patterns furnished which show size and directional lines.						
	c. Cut the nylon oxford for the front facings 2 inches in width ± 1/8 inch. Facings may be placed on rolls.						
	d. Cut the binding on the straight or bias of sufficient width to meet the requirements specified in operation 11. Binding may be placed on rolls.						
	e. Cut the tape for sleeve buttonhole tabs to meet the finished length specified in operation 10.						

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TABLE I. (cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.2.3.1 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Component marking. Mark all components of the liner to insure a uniform size throughout the liner. Any method of marking may be used except: a. Metal fastening devices. b. Sew on type tickets. c. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
4.	Labels. Sew labels 1/8 inch ± 1/16 inch from edge on all four sides to inside center (1/2 inch off center tolerance) of back as follows: (1) Size label - May abut or be caught in the line of stitching of the neck buttonhole reinforcement piece. (2) Identification label - May abut or be caught in the line of stitching of the size label. (3) Instruction label - May abut or be caught in the line of stitching of the identification label.	301	LSbj-1	10-14	50	50	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labels. (cont'd) (4) Combination size and identification label (when applicable) - Shall be positioned as in (1) above, and the instruction label shall be positioned as in (2) above. (5) Combination size, identification and instruction label (when applicable) - Shall be positioned as in (1) above.						
5.	Attach neck buttonhole reinforcement. Position reinforcement piece on inside of liner as indicated on pattern, with the top raw edge even with neck edge; turn in raw edges of sides and bottom 1/4 inch and stitch through liner 1/16 to 1/8 inch from turned edges.	301 or 401	LSd-1	10-14	50 50	50 70	
6.	Make hand openings on fronts. Place a row of stitching on both sides of cut line for hand opening. The stitching shall be 3/16 to 1/4 inch from the cut line and shall be the length of the cut line. Cut the length of the opening.	301 or 401	SSv-2	10-14	50 50	50 70	
7.	Attach facings to fronts. a. Position facings on inside of front parts respectively, with the front edges of facings even with front edges of fronts for the full length from neck edge to bottom edge. Turn in back edges of facings 3/16 to 1/4 inch and single stitch 1/16 to 1/8 inch from turned edges.	301 or 401	LSd-1	10-14	50 50	50 70	

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TABLE I. (cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Attach facings to fronts. (cont'd)						
	b. Stitch front edges of facings to front edge of front parts, 1/8 to 1/4 inch from edge.	301 or 401	SSa-1	10-14	50 50	50 70	
	- or -						
	Front edges of facings may be caught in overedge stitching of front parts (see operation 9.)						
	- or -						
	Turn in back raw edges of facings 3/16 to 1/4 inch, position on inside of front parts respectively with front edge of facings even with front edge of fronts, double stitch facings to fronts with a 1-3/8 to 1-1/2 inch gage.	301 or 401	LSd-1 and SSa-1	10-14	50 50	50 70	
8.	Join liner.						
	Finished appearance. The overedge stitched edges shall finish on the outside of liner. The bight of overedge stitching shall be 3/16 to 1/4 inch gage.						
	a. Fold pleats on undersleeves, as indicated by marks on pattern, and stitch along edge. The outside fold of pleats shall be toward the bottom.	301 or 401		10-14	50 50	50 70	
*	b. Join sleeve to undersleeve; join fronts to back at shoulders; set sleeves to armholes; join side seams and sleeve seams (one continuous operation) aligning material at sleeve end, liner bottom, and seam at underarm. All joining seams shall be stitched and overedged in one operation.	515 or 516 or 519	LSbm-3(a)	10-14	50 (chain stitching) 70 (overedge stitching)	70 70	
	- or -						

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TABLE I. (cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN/ LOOPER	THREAD COVER
8.	Join liner. (cont'd)						
	When fronts and backs are cut as one piece (see 3.3.1), join shoulder seams; join sleeve seams; set sleeves to armholes. All joining seams shall be stitched and overedged in one operation.	515 or 516 or 519	LSbm-3(a)	10-14	50 (chain stitching) 70 (overedge stitching)	70 70	
	c. Turn seam allowances of joining seams to one side and top stitch 3/16 to 1/4 inch from folded edges.	301 or 401	LSbm-3(b)	10-14	50 50	50 70	
9.	Overedge stitch raw edges.						
	Overedge stitch raw edges around fronts, neck, sleeve bottoms and bottom edge. Width of overedge stitch shall be sufficient to hold edges but not show after edges are bound.	503 or 504	SSa-1	6-12	70	70	
10.	Make and attach sleeve buttonhole tabs.						
	Finished appearance. Each liner shall have four buttonhole tabs, measuring finished and attached to sleeve, 2-7/8 ± 1/4 inches in length.						
	a. Double fold one end of each tab 1/4 to 3/8 inch and stitch or bartack 3/16 to 1/4 inch from folded edge across width of tape.	301 or 3/4 inch bartack	EFa-1	10-14 42 per bartack	50 50	50 50	
	b. Make a vertical buttonhole in the center (1/8 inch off center tolerance) of each tab, with the inside edge of eyelet 1/2 to 5/8 inch from the turned end of tab.	Buttonhole		52-56 per button-hole including tack	50	50	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make and attach sleeve buttonhole tabs. (cont'd)						
	c. Center a buttonhole tab on the inside of both sleeves over sleeve closing seams and center a second tab on the fold of the sleeves opposite the sleeve seam tab (1/4 inch off center tolerance).						
	d. With the finished end of the tabs toward the top of the sleeve, and with raw edges of the tab and bottom of sleeve even, stitch across the width of tab, 1/8 to 1/4 inch from edge. Purling shall face the inside of the sleeve when tab is extended below bottom edge of sleeve.	301 or 401	SSa-1	10-14	50 50	50 70	
11.	Bind edges of liner.						
	Finished appearance. The binding shall finish 3/8 to 5/8 inch wide. The raw ends of the sleeve buttonhole tabs, and raw edges of neck buttonhole reinforcement and facings shall be caught in the stitching of the binding.						
	a. Bind all edges of liner which include neck, fronts, bottom, sleeve bottom and hand openings. Sufficient material shall be caught at each end of hand opening to prevent binding from pulling loose.	301	BSc-1	10-14	50	50	
	b. The raw ends of binding shall be turned in and caught in the stitching; or may be turned over binding to the outside of liner 1/2 inch and tacked or bartacked.	301 or 1/2 inch bartack	SSa-1	10-14 28 per bartack	50 50	50 50	

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TABLE I. (cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Bind edges of liner. (cont'd)						
	c. Bartack binding at both ends of hand opening with 1/2 inch bartacks perpendicular to the openings.	Bartack		28 per bartack	50	50	
12.	Make buttonholes.						
	Finished appearance. The buttonholes on the fronts shall be through the facings, and 7/8 to 1 inch from front edge. The buttonholes shall be parallel to the front edges of the liner. The two neck buttonholes on the back shall be at right angles to the neck edge and shall be through the reinforcement piece, with inside edge of eyelets 1 to 1-1/4 inches from top edge of liner.						
	a. Make 12 buttonholes, five in each front and 2 at neck on back in location indicated by marks on pattern. The eyelets of the buttonholes along front edges may be up or down, i.e., on one front, the eyelet end may be up and on the other front, the eyelet end may be pointed down.	Buttonhole		52-56 per buttonhole including tack	50	50	
13.	Clean liner.						
	a. Trim and remove all thread ends, except ends of the quilting stitching on the basic material.						
	b. Remove all spots, stains, and size markings.						

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3 6 Finished measurements The liner shall conform to finished measurements shown in table II.

TABLE II. Finished measurements

Size	X-Small Inches	Small Inches	Medium Inches	Large Inches	X-Large Inches	Tolerance Inches
1/2 Chest 1/	20	22	24	26	28	± 3/4
Back length 2/	32	32-1/2	33	33-1/2	34	± 3/4
Sleeve length 3/	24	24-1/4	24-1/2	24-3/4	25	± 3/4 3/

1/ Measurement shall be taken with liner flat and front edges abutting, from the folded edge to folded edge at base of armholes.

2/ Measured from top edge at neck to bottom edge along center back.

3/ Measure the overedge side of back arm seam from top of overedge stitching at armhole to bottom of sleeve with no tension applied to seam.

3 7 Workmanship The finished liner shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4 2 1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

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4.2 2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether construction details which cannot be examined in the finished product are in accordance with specified requirements. This inspection shall include verification that the working patterns conform to the Government patterns in all respects Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process Items which cannot be corrected shall be removed from production.

4.2.3 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of one liner. The sample unit shall be one liner. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2 5 for major defects, 15.0 for major and minor A combined defects, and 40.0 for total (major, minor A and minor B combined) defects.

TABLE III. End item visual examination

Examine	Classification Major	Minor A	Minor B
Material defects and damages			
a Quilted batting			
1. Any ply having a hole, cut, tear, burn or needle chew	X		
2. Any ply having a darn or mend		X	
3. Exposed drill hole (in any ply)			X
4. One or more rows of stitching omitted in the quilting	X		
5. More than 1-1/2 inches of broken or missing stitches in the quilting			X
6 Thick or thin place	X		
7. Piecing or splicing of outer covering (either ply)	X		
b. Oxford cloth and nylon tape.			
1. Any hole, cut, or tear	X		
2. Shade bar, fine or course, filling bar (oxford cloth)		X	
3. Abrasion mark		X	
Cleanness			
a. Any spot or stain on inside			X
b Thread ends not trimmed or loose threads not removed on major portions of liner, except thread ends of quilting stitching on basic material			X
c. One or more size markings not removed			X

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Component and assembly			
a. Any component part or required operation omitted (unless otherwise classified herein)	X		
b. Any operation not performed as specified (unless otherwise classified herein)			X
Cutting			
Any component part not cut in accordance with directional lines indicated on pattern or not cut in accordance with specified requirements	X		
Seams and stitching			
a Accuracy of seaming:			
1. Seam twisted, puckered, or pleated			X
NOTE Nylon fabric has a tendency to pucker at joining seams and at stitchings. The puckering when not so excessive as to seriously affect appearance shall not be classified as a defect.			
2. Part of liner caught in any unrelated operation or stitching	X		
3. End of seam or stitching produced with 301 stitch type, when not caught in other seams or stitching, backtacked less than 1/4 inch		X	
4. Stitching overlapping end of thread break less than 1/2 inch			X
5 Various shades of thread used on outside of same liner		X	
6 Ends of a continuous line of stitching not overlapped or overlapped less than 1/2 inch			X
b. Gage of stitching and seam allowances			
1 Irregular, i.e., unevenly gaged or corresponding stitchings not uniformly gaged (to be scored only when the condition exists on major portion of seam)		X	

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
2. Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			X
3. Edge or raised stitching sewn too close to edge resulting in damage to cloth	X		
4. Open seams: (not applicable to quilting stitching)			
Any seam open by:			
- more than 1 inch	X		
- more than 1/4 inch up to 1 inch inclusive		X	
- up to 1/4 inch inclusive			X
NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On double stitched seams, a seam is classified as open when either one side or both sides of seam are open.			
5. Raw edges (excluding overedge stitching and end of binding, when not turned under):			
- more than 1 inch	X		
- more than 1/4 inch up to 1 inch inclusive		X	
- 1/8 inch up to 1/4 inch inclusive			X
6. Any raw edge caused by omission of overedge stitching for a distance of 3 inches or more		X	
NOTE: Raw edges are classified as such when they occur at places where edge is required to be bound, turned under or overedge stitched. Raw edges not securely caught in stitching shall be classified as open seams.			

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
7 Run-off:			
(a) Joining seams - When resulting in an open seam use "Open seam" classification			
(b) Edge or raised stitching (when not resulting in open seam):			
- 1/4 inch up to 1/2 inch inclusive			X
- more than 1/2 inch		X	
8. Seam and stitch type:			
(a) Stitch or stitch type not as specified		X	
(b) Overedge stitched edge on joining seams not finishing on outside of liner		X	
(c) Looper thread of 401 stitch type on joining seam finishing on inside of seam (LSbm-3 seam type only)			X
9. Stitch tension:			
(a) Loose tension resulting in a loose seam		X	
(b) Loose tension on raised or edge stitching			X
(c) Tight tension (stitches break when strain is applied to seam or stitching)		X	
NOTE: Puckering is evidence of tight tension. When puckering is evident, the seam shall be tested by exerting pull in lengthwise direction of seam.			
c. Stitches per inch (to be scored only when the condition exists on the major portion of seam; except labels):			
1. Less than minimum specified:			
- one or two stitches			X
- three or more stitches	X		
2. One or more stitches less than minimum specified on one or more labels		X	
3. One or more stitches more than specified			X

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Bartacks			
a Any bartack omitted		X	
b. Loose stitching, incomplete or broken			X
c Any bartack not in specified location			X
d Length or width not as specified			X
Buttonholes (construction)			
a. Stitching not caught on fabric, causing stitching to pull away from fabric	X		
b. Wrong type, incomplete stitching or uncut buttonhole		X	
c. One or more broken stitches in:			
- one buttonhole		X	
- two or more buttonholes	X		
d More than two skipped stitches in:			
- one buttonhole			X
- two or more buttonholes		X	
e Gimp omitted in one or more buttonholes			X
f. Buttonhole less than 3/4 inch or more than 7/8 inch long		X	
g Purling on outside of liner			X
h. End stitched with less than 4 stitches per bar end (see 3.6.2):			
- one buttonhole		X	
- two or more buttonholes	X		
Sleeves			
Fold of pleat toward shoulder			X
Binding			
a. Not caught on fabric, i e., can easily be pulled away from edge of fabric	X		
b. End of binding not turned over the binding to outside of liner and tacked, or not turned under and caught in the stitching			X
c Binding finishing less than 3/8 or more than 5/8 inch wide			X
d Binding stretched, i.e., edges have a wavy appearance			X

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Front facings			
a. Stitched on outside of liner	X		
b. Front, top or bottom edge not completely caught in binding stitching		X	
c. Tight, short, or twisted, causing fullness or twist on fronts			X
d. Back edge not turned under and stitched		X	
Front buttonholes (position)			
a. Unequally spaced by more than 1/2 inch		X	
b. Front buttonholes less than 7/8 inch, or more than 1 inch from front edge, measured from inside edge of eyelet		X	
c. Not parallel to front edge			X
Neck buttonholes (position)			
a. Inside edge of eyelet less than 1 inch or more than 1-1/4 inches from top edge of liner at back			X
b. Eyelet end not toward neck edge of liner or positioned horizontally		X	
c One or more not through reinforcement piece		X	
Sleeve buttonhole tabs			
a. One or more omitted, or not caught in binding stitching	X		
b. Less than 2-5/8 or more than 3-1/8 inches long			X
c. Placed more than 1/4 inch from center of sleeve seam or more than 1/4 inch from center of fold opposite sleeve seam			X
d. Placed on angle, not permitting tab to be in vertical position			X
e. One or more tabs not positioned on the inside of liner		X	
f. Inside edge of eyelet less than 1/2 or more than 5/8 inch from bottom edge			X
g. Buttonhole off center with tab width by more than 1/8 inch			X
h Bottom hem of tab less than 1/4 or more than 3/8 inch in length			X
i. Purling of buttonhole not toward the inside of liner when tab is extended below bottom edge of sleeve			X

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TABLE III. End item visual examination (cont'd)

Examine	Classification Major	Minor A	Minor B
Labels			
a Size, instruction or combination label (when applicable) missing, incorrect or illegible	X		
b Identification label missing, incorrect or illegible		X	
c. Any label not stitched on four sides			X
d. Any label off center on back by more than 1/2 inch			X
e Not positioned as specified		X	

4 2 4 End item dimensional examination. The end item shall be examined for conformance to the finished dimensions specified in table II Any dimension not within the specified tolerance shall be classified as a defect The lot size shall be expressed in units of one liner The sample unit shall be one liner The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4 2 5 Packaging inspection An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2 5.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence,or method of application.
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number of bundles per container is more or less than required. Number of liners per bundle is more or less than required. 1/

1/ For this defect, one bundle from each container in the sample shall be examined.

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4.2.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below The sample unit shall be one palletized unit load fully packaged The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6 5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A Each liner shall be laid flat, front up with front edges abutting. The sleeves shall be folded at right angles across the chest. For the larger sizes, it may be necessary to adjust the sides in order to attain the desired width of approximately 23 inches. The liner shall then be folded in half the length. The completely folded liner shall measure approximately 23 by 14-1/2 inches. Five folded liners of one size only shall be neatly stacked in a bundle and securely tied with cotton tape or twine approximately 4 inches in from each end.

5.1.2 Commercial packaging. The liners shall be preserved in accordance with ASTM D-3951.

5 2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2 1 Level A packing Fifteen liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-

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resistant, variety DW, grade V15c of PPP-B-636 Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth Approximate dimensions are furnished as a guide only. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4 2 5 Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes of liners may be packed in the same container.

5.2 2 Level B packing Fifteen liners of one size only, preserved as specified in 5 1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636 Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth Approximate dimensions are furnished as a guide only. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4 2 5. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes of liners may be packed in the same container.

5.2.2 1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.5.

5.2.3 Commercial packing. Liners, preserved as specified in 5 1, shall be packed in accordance with ASTM D-3951

5.3 Palletization. When specified (see 6.2), liners packed as specified in 5 2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147 Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O and P. Pallet patterns shall be number 3 in accordance with the appendix of MIL-STD-147 Inter-locking of loads shall be effected by reversing the pattern of each course

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5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

5.4.1 Labels, mixes sizes. Each shipping container, packed with mixed NSN's, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5- by 4-inches, with the words "MIXED NSN's" plainly stamped or printed thereon, and under these words shall be legibly printed or stamped the correct quantity and sizes container therein.

6. NOTES

6.1 Intended use. The liner is intended for wear by military personnel of the Department of Defense as a component of the desert combat uniform in conjunction with the Parka, Night Camouflage, Desert.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Size required (see 1.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 Samples. For access to samples address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Marginal notations. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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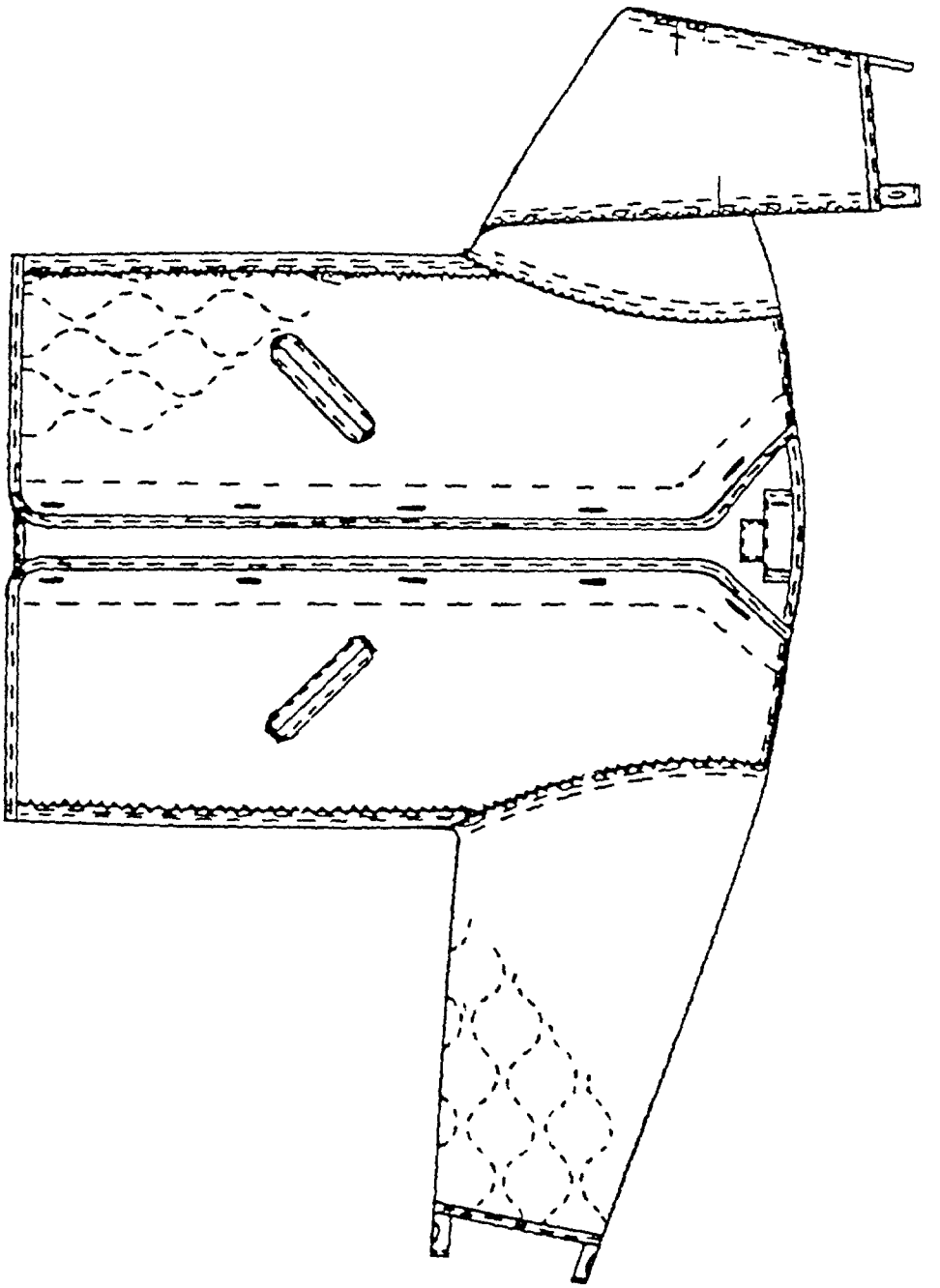
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FIG. 1 LINER, NIGHT CAMOUFLAGE PARKA: DESERT



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-L-44089A	Liner, Night Camouflage Parka: Desert

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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